Data Analytics, Data Science, Artificial Intelligence, and Natural Language Processing For Local Business Review Analysis

Sentiment and Rating Prediction from User Feedback

A Research Report

Project: Restaurant Review Sentiment & Aspect Analyzer

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1. Abstract

Online customer reviews are one of the most important factors that influence customer buying decisions and a business's reputation. This report presents a Local Business Review Analyzer developed for the restaurant industry by combining Data Analytics, Data Science, Artificial Intelligence (AI), and Natural Language Processing (NLP) to convert unstructured customer reviews into meaningful insights. A dataset of 577 cleaned reviews from ten restaurants in Bengaluru was manually collected and carefully selected instead of using automated web scraping. This approach ensured compliance with platform terms of service while also providing a balanced collection of reviews across different rating levels.

The review processing pipeline includes text cleaning, tokenization, stopword removal, lemmatization, and a custom negation-handling method to improve text understanding. The system then uses TF-IDF features based on word unigrams, bigrams, and character n-grams, along with lexicon-based sentiment features, to represent the review text. A regularized Logistic Regression model, optimized using grid search and stratified cross-validation, is used to predict the review's star rating directly. The predicted rating is further used to determine the overall sentiment polarity and an expected rating score.

For deeper analysis, Latent Dirichlet Allocation (LDA) is applied to discover hidden topics discussed in customer reviews. In addition, a rule-based aspect detection method identifies important review categories such as Food, Service, Ambience, and Price. The results are displayed through an interactive Streamlit dashboard that provides business-wise performance metrics, rating distributions, aspect analysis, topic visualization, and word clouds. This report explains the complete methodology, compares the selected modelling approach with other techniques discussed in previous research, and highlights the business value of the system along with possible future improvements.

2. Introduction

2.1 Importance of Online Customer Reviews

Customer reviews on platforms such as Google Maps and Zomato have become one of the main factors influencing new customers and an important source of feedback for business owners. A regular flow of reviews affects customer visits, brand image, and even search rankings on online discovery platforms. For small and local businesses, even a few negative reviews can significantly reduce revenue, while positive reviews help build customer trust and encourage recommendations. However, the large number and unstructured format of reviews make manual analysis difficult, creating a clear need for automated review analysis.

2.2 Role of Data Analytics, Data Science, AI, and NLP

Data Analytics and Data Science provide the statistical and computational methods for analysing review counts, rating patterns, and comparisons between businesses. Artificial Intelligence (AI) and Natural Language Processing (NLP) build on this by analysing the review text, allowing computers to identify sentiment, predict star ratings from written reviews, and discover hidden topics without predefined labels. Together, these fields turn raw customer feedback into clear, measurable, and useful business insights.

2.3 Problem Statement

Local businesses often do not have the right tools to properly analyse the sentiment, key quality factors, and main topics found in customer reviews. Star ratings alone give only limited information because two reviews with the same rating can describe completely different experiences. Also, the exact reason for customer dissatisfaction, such as food quality, service speed, ambience, or pricing, is not always clear. This project solves this problem by converting manually collected review text into (i) sentiment classification, (ii) predicted or expected ratings, and (iii) understandable topics and aspects, without using any internal business data.

2.4 Objectives and Scope

- Create a manually collected and rating-balanced dataset of restaurant reviews suitable for supervised learning.
- Develop an NLP preprocessing pipeline that includes text cleaning, tokenization, stopword removal, lemmatization, and negation handling.
- Train and evaluate a supervised machine learning model to predict star ratings and classify sentiment.
- Use LDA topic modelling and rule-based aspect tagging to identify hidden topics and important aspects in customer reviews.
- Present the results through an interactive Streamlit dashboard that is easy for non-technical business users to understand.

The scope of this project is limited to English-language, text-based restaurant reviews collected from a single city and a single platform. It does not include multilingual reviews or reviews containing images or videos.

3. Literature Review

Sentiment analysis and review mining have been important research areas for more than twenty years. Early research by Pang and Lee (2008) treated sentiment classification as a text classification problem. They applied machine learning algorithms such as Naive Bayes, Maximum Entropy, and Support Vector Machines (SVM) to identify the sentiment of movie reviews, creating benchmark methods that are still widely used today. Later, Liu (2012) expanded this work by presenting a complete framework for opinion mining. He explained the differences between document-level, sentence-level, and aspect-level sentiment analysis, with aspect-level analysis forming the basis of the aspect-tagging approach used in this project.

Hu and Liu (2004) were among the first researchers to introduce aspect-based opinion summarization. Their method extracted product features and the opinions related to them from customer reviews using frequent-itemset mining. This idea is closely related to the rule-based Food, Service, Ambience, and Price tagging method used in this project. Blei, Ng, and Jordan (2003) introduced Latent Dirichlet Allocation (LDA), a probabilistic topic modelling technique that is directly used in this project to discover hidden topics from reviews without requiring manual labels.

As research progressed, traditional machine learning methods based on feature engineering were gradually complemented by distributed word representations and deep learning models. Mikolov et al. (2013) introduced Word2Vec, which represents words as vectors and captures semantic relationships beyond simple bag-of-words counts. Later, Devlin et al. (2019) introduced BERT, a transformer-based language model that achieved state-of-the-art performance in many NLP tasks, including sentiment analysis, by fine-tuning a pretrained model on labelled datasets.

At the same time, ensemble learning and gradient-boosting methods became popular for structured and semi-structured prediction tasks. Breiman (2001) introduced Random Forests, while Chen and Guestrin (2016) introduced XGBoost. Both algorithms have been widely used for rating prediction using review-based features such as TF-IDF scores, review length, and sentiment lexicon features. Comparative studies, including Fang and Zhan (2015), evaluated algorithms such as Naive Bayes, SVM, and Logistic Regression on Amazon review datasets. Their results showed that well-regularized linear models, especially when combined with n-gram and lexicon features, can perform as well as or even better than more complex models on small and medium-sized domain-specific datasets.

Rating prediction has been approached in two main ways: as a regression problem using models such as Linear Regression and Random Forest Regression, or as an ordinal classification problem where a multiclass classifier predicts class probabilities that are converted into an expected rating. The second approach is the one adopted in this project. Topic modelling has also advanced from LDA to newer neural approaches such as BERTopic (Grootendorst, 2022). However, LDA, combined with coherence-based topic selection (Röder, Both, and Hinneburg, 2015), is still considered a reliable and easy-to-interpret baseline for small and medium-sized academic datasets like the one used in this study.

Overall, previous research supports two important conclusions that are directly relevant to this project. First, classical machine learning models with proper regularization and carefully designed features, such as TF-IDF n-grams, sentiment lexicon features, and negation handling, remain effective and competitive for small, domain-specific review datasets. In comparison, deep transformer models generally require much larger labelled datasets to achieve better performance. Second, LDA continues to be a dependable unsupervised method for discovering hidden topics when combined with systematic coherence evaluation.

4. Methodology

4.1 Dataset Description

The dataset consists of manually collected customer reviews for ten restaurants located in Bengaluru (including MTR, Vidyarthi Bhavan, Koko, Nagarjuna, Empire Restaurant, Meghana Foods, Paragon, Truffles, and CTR Shri Sagar). Each record contains the business name, the full review text, and the associated 1–5 star rating. After cleaning and de-duplication, the working corpus contains 577 reviews across these ten businesses.

4.2 Data Collection Process

Reviews were curated manually from each restaurant's public review page rather than through automated scraping, a deliberate choice documented in the project's data-collection notes. Automated scraping was avoided for three reasons: it would risk violating platform terms of service, it is vulnerable to anti-bot defences, and it tends to return reviews skewed toward the platform's default ordering, which over-represents 4–5 star reviews. Manual, stratified collection allowed reviews to be deliberately drawn across all rating levels per restaurant. In the final cleaned dataset the rating distribution is 1 star rating: 57, 2 star rating: 36, 3 star rating: 52, 4 star rating: 144, and 5 star rating: 288 reviews — still skewed toward positive ratings, reflecting the natural distribution of restaurant reviews even under stratified sampling, and treated as a known limitation rather than a fully balanced sample.

4.3 Data Preprocessing and Feature Engineering

- Removal of duplicate rows and whitespace normalisation on business names (a trailing-space inconsistency was identified and corrected).
- Missing-value imputation for unlabelled businesses.
- Text normalisation: lower-casing, punctuation stripping, tokenization, English stopword removal (NLTK), and WordNet lemmatization.
- A custom negation-tagging scheme that prefixes tokens following a negator word (e.g., "not", "never", "without") with a neg_ marker within a bounded window, so that phrases such as "not good" are distinguished from "good" during vectorization.
- Construction of a sentiment lexicon (curated positive/negative restaurant-domain word lists) used both as an interpretability aid and as an additional model feature.

4.4 Model Selection Rationale

Since the dataset was moderate in size and specific to one domain (577 labelled reviews), a regularized **Logistic Regression** model was chosen instead of deep transformer-based models such as **BERT**, which usually need much larger labelled datasets and higher computing resources to achieve better performance. Logistic Regression was combined with multiple text representation methods instead of using only one vectorizer. These included word unigrams and bigrams to capture individual words and short phrases, character n-grams to handle spelling mistakes and informal review language, and lexicon-based features to improve sentiment detection. The model's hyperparameter (regularization strength **C**) was optimized using grid search with stratified 4-fold cross-validation, while maximizing the macro-F1 score to handle class imbalance across the five star-rating levels.

5. NLP and Machine Learning Techniques

5.1 TF-IDF and Text Vectorization

Term Frequency–Inverse Document Frequency (**TF-IDF**) was used to convert the cleaned review text into numerical feature vectors. This method reduces the importance of very common words while giving more importance to words that are unique and meaningful in each review. Three TF-IDF representations were combined using a feature union: word unigrams (**min_df = 1**), word bigrams (**min_df = 2**) to capture short phrases such as "cold food" and "friendly staff", and character n-grams (3–5 characters) to make the model more robust to spelling mistakes and informal language commonly found in customer reviews.

5.2 Sentiment Analysis

In this project, sentiment classification is not performed using a separate model. Instead, it is obtained from the rating prediction model. The classifier predicts the probability of each star rating, and these predicted ratings are then converted into **positive**, **neutral**, or **negative** sentiment using a fixed rating-to-sentiment mapping. This approach is based on research showing that, for restaurant reviews, sentiment and star ratings are closely related. Therefore, using a single well-optimized classifier is simpler and more efficient than building separate models for sentiment classification and rating prediction. Although **Naive Bayes** and transformer-based models such as **BERT** are widely used for sentiment analysis, they were not selected for this project because of the moderate dataset size and the strong performance achieved by a regularized Logistic Regression model using a combination of carefully designed text features.

5.3 Rating Prediction

Rating prediction in this project is performed by converting an ordinal classification problem into a regression-style output. The trained Logistic Regression model produces probability scores for each of the five rating categories, and an expected rating value is calculated by taking the weighted average of these probabilities and their corresponding rating values. The model was tested using 5-fold cross-validation, with evaluation based on Rating Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score. Standard classification metrics such as precision, recall, and F1-score were also used to evaluate the direct 1–5 star rating prediction task. Models such as Linear Regression, Random Forest, and XGBoost are commonly used alternatives for rating prediction and are discussed as comparison methods. However, the selected approach was preferred because it combines sentiment analysis and rating prediction into a single, simpler, and easier-to-maintain model.

5.4 Topic Modeling

Latent Dirichlet Allocation (LDA), implemented using **Gensim**, was applied to the cleaned and stopword-removed review dataset to identify hidden topics without using predefined categories. Different topic numbers (**k = 3, 4, 5, and 6**) were tested, and their **c_v coherence scores** were compared to select the topic number that provided the best results. The identified topics were automatically given meaningful labels using the project's rule-based aspect tagger, which analysed the most important words in each topic and mapped them to categories such as **Food, Service, Ambience, and Price**. Additionally, a rule-based aspect tagger was applied directly to individual reviews to identify their main aspect and assign a confidence score. This provided both a dataset-level view through topic modelling and a review-level view through aspect analysis.

5.5 Comparison of Techniques

Technique	Role in Literature	Status in this project
Logistic Regression + TF-IDF ensemble	Strong, interpretable baseline for text classification	Implemented (core model)
Naive Bayes	Classic, fast text-classification baseline	Discussed in literature; not implemented
BERT (Transformer)	State-of-the-art contextual sentiment classification	Discussed in literature; not implemented (dataset size)
Linear Regression / Random Forest / XGBoost	Common rating-prediction regressors	Discussed in literature; rating derived from LR class probabilities instead
LDA Topic Modeling	Unsupervised latent theme discovery	Implemented, with coherence-based k selection
Rule-based Aspect Tagging	Aspect-based opinion mining	Implemented (Food / Service / Ambience / Price)

6. Tools & Technologies

- Python — core programming language for the entire pipeline.
- Pandas and NumPy — data loading, cleaning, merging, and numerical operations.
- Scikit-learn — TF-IDF vectorization, Logistic Regression, GridSearchCV, cross-validation, and evaluation metrics.
- NLTK — stopword removal, WordNet lemmatization.
- Gensim — LDA topic modelling and coherence scoring.
- Matplotlib, Seaborn, and WordCloud — rating distribution, review-length distribution, per-business review counts, and word-cloud visualisations.
- Jupyter Notebook — exploratory data analysis, model development, and evaluation workflows.
- Streamlit — the interactive dashboard exposing per-business metrics, rating/aspect breakdowns, and word clouds to non-technical users.
- Git and GitHub — version control and project source management.

This toolchain reflects a lightweight, reproducible, and fully open-source stack suitable for an academic-scale, single-researcher project.

7. Conclusion

7.1 Summary of Project Workflow

This project delivers an end-to-end pipeline: manual, rating-stratified review collection; cleaning and NLP preprocessing with negation handling; a tuned Logistic Regression model trained on an ensemble of TF-IDF and lexicon features to jointly predict star rating and derive sentiment; LDA-based topic modelling with coherence-optimised topic count; rule-based aspect tagging; and an interactive Streamlit dashboard for exploration by business owners or analysts.

7.2 Business Insights from Customer Reviews

The pipeline enables a business to see, at a glance, its average rating and sentiment mix, which aspect of the experience (food, service, ambience, or price) is most frequently discussed, and the underlying topics driving both praise and complaints — insight that a raw star-rating average alone cannot provide.

7.3 Expected Outcomes

The system is expected to provide restaurant owners and analysts with a low-cost, interpretable tool for monitoring review sentiment and identifying specific, actionable improvement areas, without requiring access to a review platform's internal API or paid analytics tooling.

7.4 Future Scope

- Expanding the corpus across more cities, cuisines, and languages to improve generalisability.
- Evaluating transformer-based models (e.g., fine-tuned BERT) once a substantially larger labelled corpus is available.
- Adding ensemble regressors (Random Forest, XGBoost) as a benchmark comparison against the current Logistic-Regression-derived rating estimate.
- Automating periodic re-scraping (with platform permission or official APIs) to keep the dashboard continuously updated.

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